TONER-1 Pro

15,191,712 parameters
int8 weights, one scale per row
Inference runs inside the PostScript interpreter

State of the art among language models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, there was a big hall. In the hall, there were many people who played and danced. They were so happy! One day, a little girl named Lily went to the hall to play. She was excited to be independent and play by herself.

Lily saw a big dog near the room. The dog was not nice. He barked at Lily and bit her. Lily felt sad and scared. She started to cry.

A kind man saw Lily crying and came to help. He talked to the dog and said, “Please be kind to Lily.” The dog stopped barking and did not bite Lily. Instead, the dog played with Lily and they had fun together. Lily was happy and learned that being independent is good.

TOKEN STREAM

The first 142 of 168 sampled tokens, one box per vocabulary piece. · marks a space, ¶ a newline.

·Once	·upon	·a	·time	,	·there	·was	·a	·big	·hall	.	·In	·the	·hall	,	·there		
·were	·many	·people	·who	·played	·and	·dan	ced	.	·They	·were	·so	·happy	!				
·One	·day	,	·a	·little	·girl	·named	·L	ily	·went	·to	·the	·hall	·to	·play			
.	·She	·was	·excited	·to	·be	·independent	·and	·play	·by	·herself	.	¶	L				
ily	·saw	·a	·big	·dog	·near	·the	·room	.	·The	·dog	·was	·not	·nice	.	·He		
·b	ark	ed	·at	·L	ily	·and	·bit	·her	.	·L	ily	·felt	·sad	·and	·sc	ared	.
·She	·started	·to	·cry	.	¶	A	·kind	·man	·saw	·L	ily	·cry	ing	·and	·came		
·to	·help	.	·He	·talked	·to	·the	·dog	·and	·said	,	·"	Please	·be	·kind	·to		
·L	ily	."	·The	·dog	·stopped	·b	ark	ing	·and	·did	·not	·b	ite	·L	ily	.	

HOW THIS PAGE WAS MADE

1 HOST
Sends the print job. Performs no arithmetic.

→

2 INTERPRETER
Parses the int8 weights, then runs the forward pass once per token.

→

3 PAGE
The same job typesets the story and calls showpage.

TOKENS	SECONDS	TOKENS / SEC	SEED	TEMPERATURE
168	**113.2**	**1.48**	**42**	**0.8**

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
MODEL	stories15M (Llama-2 architecture) · dim 288 · 6 layers · 6 query / 6 KV heads · 32,000-token vocab
STOPPED BY	End-of-story token, after 168 tokens

Host compute: 0 FLOPs. Inference cost includes toner.

Every word above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seed on a different interpreter tells a different story.